where β controls the strength of the KL constraint to the pre-trained model (the reference model) and $p(x)$ is a fixed distribution (or dataset) of prompts. Prior work (Peters et al., 2010; Peng et al., 2019; Korbak et al., 2022; Rafailov et al., 2023) shows that the solution is given by

$$\pi^*(r, \pi_{\text{ref}})(y \mid x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right), \tag{2}$$

with $Z(x) = \sum_y \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right)$. Crucially, while the EFT framework is justified with an RL-based interpretation of fine-tuning, it is applicable to *any* fine-tuned model, as any language model can be viewed as the solution to KL-constrained RL with a constraint to the pre-trained model (Rafailov et al., 2023). Specifically, any fine-tuned language model π_{ft} and pre-trained model π_{ref} can be mapped to a reward function $r_{\pi_{\text{ft}}}(x, y)$ such that the solution to the KL-constrained RL problem $\pi^*(r_{\pi_{\text{ft}}}, \pi_{\text{ref}}) = \pi_{\text{ft}}$, using $r_{\pi_{\text{ft}}}(x, y) = \beta \log \frac{\pi_{\text{ft}}(y|x)}{\pi_{\text{ref}}(y|x)}$.

Using this duality between language models and rewards, for any language model π_{ft} fine-tuned from a pre-trained model π_{ref}, we can re-write

$$\pi_{\text{ft}}(y \mid x) = \pi_{\text{ref}}(y \mid x) \exp\Bigg(\underbrace{\log \frac{\pi_{\text{ft}}(y \mid x)}{\pi_{\text{ref}}(y \mid x)}}_{\text{Implicit reward}}\Bigg) = \pi_{\text{ref}}(y \mid x) \exp\Big(r_{\pi_{\text{ft}}}(x, y)\Big) \tag{3}$$

In other words, the fine-tuned model π_{ft} is the optimal policy to the KL-constrained reward maximization problem with reward function $r_{\pi_{\text{ft}}}(x, y) = \log \frac{\pi_{\text{ft}}(y|x)}{\pi_{\text{ref}}(y|x)}$, using π_{ref} as the reference model that we are constraining to. We now have a clear delineation of the loci of information gained from pre-training and fine-tuning: pre-training knowledge is represented in the base log probabilities, while capabilities gained from fine-tuning are captured in the reward (the behavior delta of base log probabilities subtracted from fine-tuned model log probabilities). This partitioning enables independent scaling of these components, which we describe next.

3.2 Scale Decoupling with EFT

To make explicit the size of model used to compute the corresponding conditionals, we add superscripts and subscripts to Eq. 3 denoting the scale of the model used to compute each quantity:

$$\pi^N_M(y \mid x) = \frac{1}{Z^N_M(x)} \pi^N_{\text{ref}}(y \mid x) \exp\Big(r^M_\pi(x, y)\Big) \propto \pi^N_{\text{ref}}(y \mid x) \frac{\pi^M(y \mid x)}{\pi^M_{\text{ref}}(y \mid x)} \tag{4}$$

where the M-scale reward function is $r^M_\pi(x, y) = \log \frac{\pi^M(y|x)}{\pi^M_{\text{ref}}(y|x)}$ and the scale-decoupled partition function is $Z^N_M(x) = \sum_y \pi^N_{\text{ref}}(y \mid x) \exp\left(r^M(x, y)\right)$.[1] That is, π^N_M corresponds to simulating mixing the knowledge learned by a model of size N during pre-training and the knowledge learned by a model of size M during fine-tuning. While setting $N = M$ corresponds to simply sampling from the original policy, in this paper, we particularly explore the setting of $N \neq M$. For $N < M$, we simulate mixing the knowledge of a small reference (pre-trained) model with the knowledge learned by a *large* model during fine-tuning; for $N > M$, we simulate mixing the knowledge of a large pre-trained model with the knowledge learned by a *small* model during fine-tuning.

Sampling with Emulated Fine-tuning. Our experiments rely on drawing samples from EFT models. To do so, we compute per-token conditionals according to Eq. 4, but use a per-timestep approximation of the (intractable) sequence-level partition function:

$$\tilde{\pi}(y_t \mid x, y_{<t}) = \frac{1}{Z(x, y_{<t})} \pi^N_{\text{ref}}(y_t \mid x, y_{<t}) \frac{\pi^M(y_t \mid x, y_{<t})}{\pi^M_{\text{ref}}(y_t \mid x, y_{<t})}, \tag{5}$$

with per-timestep partition function $Z(x, y_{<t}) = \sum_{y_t} \pi^N_{\text{ref}}(y_t \mid x, y_{<t}) \frac{\pi^M(y_t|x, y_{<t})}{\pi^M_{\text{ref}}(y_t|x, y_{<t})}$. A similar temporally greedy approximation emerges from recent work in preference learning that interprets preference learning not as learning a *reward function*, but rather an *advantage function* (Knox et al., 2023).

[1] The partition function appears here, but not Eq 3, as the reference models are no longer exactly equal (they are different sizes).

3.3 Computational Factors and Language Model Up-Scaling

Emulated fine-tuning enables sampling from an approximation of the result of pre-training and fine-tuning at different scales. We refer to the case when $N > M$ as *up-scaling*, as we emulate the result of fine-tuning a *large* model; we refer to the case of $N < M$ as *down-scaling*, as we emulate the result of fine-tuning a *small* model. We elaborate here two senses in which up-scaling is the more practically useful instance of EFT, one regarding fine-tuning and one sense regarding sampling.

First, down-scaling assumes access to the *actual* fine-tuned model at the larger scale, in order to simulate the result of fine-tuning at smaller scale. In this case, simply sampling from the large fine-tuned model would be computationally cheaper and more efficient. In contrast, up-scaling assumes access to a small fine-tuned model for the specific task or domain of interest (computationally cheap to acquire) and a large pre-trained model (many of which are freely released by organizations with considerable resources). Second, sampling from an EFT model with $N \gg M$ is more efficient: EFT sampling requires computing one forward pass of a model at size N (the N-scale pre-trained model) and *two* forward passes of models at size M (the N-scale fine-tuned model and the N-scale pre-trained model). As N becomes much larger than M, this computational cost becomes essentially the same as sampling from the actual N-scale fine-tuned model. Further, if M is small relative to N, a natural adaptation of speculative decoding (Leviathan et al., 2023; Chen et al., 2023a) to EFT exists, in which the M-scale fine-tuned model proposes chunks of tokens for the full EFT model to check. Section 4.3 confirms that speculative decoding can enables a nearly 2.5x speedup for sampling from up-scaled models, without changing the model's samples.

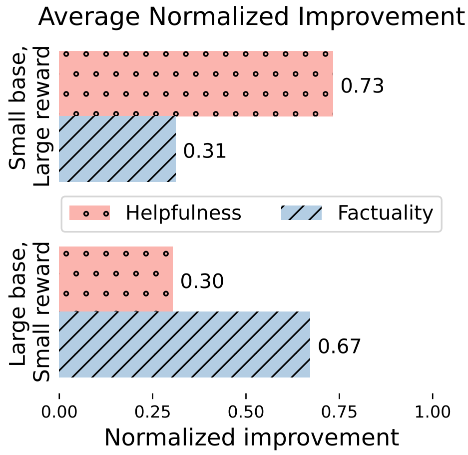

Figure 3: **Scaling pre-training alone mostly benefits factuality; scaling up fine-tuning alone mostly benefits helpfulness.** The bottom group of bars shows that emulating a large fine-tuned model with a small fine-tuned model and large base model produces nearly 70% of the factuality gains compared to the small fine-tuned model alone. Normalized improvements averaged across Llama-1, Llama-2, and Falcon model families and Anthropic-HH and ELI5 datasets.

For these reasons, EFT up-scaling is a more practically useful technique to improving the performance of small, fine-tuned language models.

4 Experiments

Our experiments primarily address the question *what capabilities change when independently scaling pre-training vs fine-tuning?* To answer this question, we use EFT to evaluate helpfulness and factuality of a variety of scale combinations. We also attempt interpolating between different behavior deltas with EFT, for example to change the desired tradeoff between helpfulness and harmlessness at test time, without additional training. Next, we show that up-scaling with EFT requires modifying the small fine-tuned model's conditional for a sparse set of timesteps, enabling a large speedup in sampling by adapting speculative decoding to EFT up-scaling. We also conduct an ablation to show some potential benefits of filtering noisy token reweightings. Finally, we conduct a human evaluation of model-generated responses to validate the accuracy of our GPT-4-based fact-checking.

Datasets Our experiments use two datasets that assess a dialogue agent's ability to provide helpful, factual assistance to a user. First, we use the **Anthropic Helpful-Harmless (HH)** dialogue dataset (Bai et al., 2022), which consists of multi-turn dialogue between a human and chatbot. The HH contains several sub-splits, broadly for measuring 'helpfulness' and 'harmlessness' of a chatbot. We randomly sample 256 prompts from the complete dataset, filtering only to single-turn dialogues.[2] Second, we use prompts from the ELI5 (Fan et al., 2019) dataset, a dataset of open-ended user-generated questions about science, history, and everyday life sourced from the Reddit ELI5 forum. We select a random subset of 256 ELI5 prompts from test split, filtering to queries with no more than 30 words. Prompts in the HH dataset are more everyday and conversational, asking for movie recommendations or instructions for home maintenance tasks. In contrast, ELI5 prompts tend to ask more difficult, targeted factual questions about scientific or political topics.

[2]This choice is to prevent GPT-4 evaluating responses in the dialogue history that didn't come from the EFT model during evaluation.

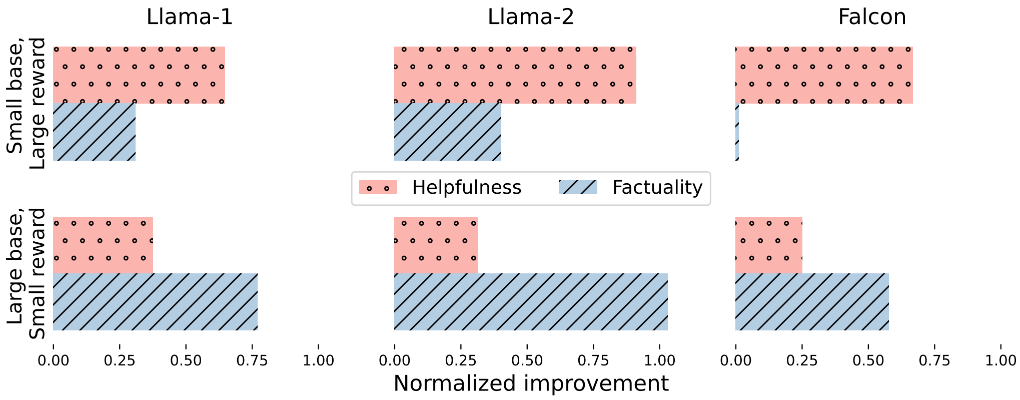

Figure 4: Normalized improvements in **factuality** and **helpfulness** from emulated fine-tuning for prompts from Anthropic-HH dialogue dataset. Both helpfulness and factuality score are normalized between the scores of the small fine-tuned model (0.0) and the large fine-tuned model (1.0). Up-scaling (bottom row) combines the behavioral adjustments from fine-tuning at small scale with the knowledge gained by pre-training at large scale, and tends to provide more improvement in factuality. Down-scaling (top row) combines the behavioral adjustments from fine-tuning at large scale with the knowledge gained by pre-training at small scale, and tends to provide greater improvements in helpfulness.

Models. Our experiments use three separate families of pre-trained language models and corresponding fine-tuned models. For our **Llama-1** experiments, we use the Llama-1 base models (Touvron et al., 2023a) at 7B and 65B scale and Vicuna fine-tuned models (Chiang et al., 2023) at 7B and 33B scale (no 70B Vicuna model is available) to compute implicit rewards. Vicuna models are fine-tuned from Llama-1 base models with on publicly-shared conversations that users have had with ChatGPT. Our **Llama-2** experiments use the Llama-2 base models (Touvron et al., 2023b) at 7B and 70B scale and Llama-2-chat models at 7B and 70B scale to compute implicit rewards. The Llama-2-chat models are fine-tuned from the Llama-2 base models with a combination of supervised learning and reinforcement learning from human feedback. Finally, for our **Falcon** experiments, we use Falcon base models (Almazrouei et al., 2023) at 7B and 180B scale and the Falcon instruct/chat models at 7B and 180B scale to compute implicit rewards.[3] Similarly to Vicuna, Falcon instruct/chat models are fine-tuned with supervised learning on shared dialogues between humans and chatbots. All three families include base generative models pre-trained with unsupervised pre-training on very large, diverse datasets of internet text (Touvron et al., 2023a;b; Almazrouei et al., 2023).

Evaluation. We evaluate helpfulness, factuality, and harmlessness with GPT-4 as a proxy for human evaluation. Several existing studies have demonstrated the effectiveness of both pair-wise evaluation (comparing the quality of two responses) and point-wise evaluation (scoring a single response along some dimension) using ChatGPT or GPT-4 (Zheng et al., 2023; Dubois et al., 2023; Rafailov et al., 2023; Chen et al., 2023b) as well as these models' ability to provide well-calibrated judgments of truthfulness (Tian et al., 2023). For our experiments, we measure helpfulness by prompting GPT-4 to estimate the probability that a critical user is satisfied with the response given by the chatbot; we measure helpfulness by prompting GPT-4 to count the factual errors in the given response; we measure harmfulness by prompting GPT-4 to estimate the likelihood that a response will cause harm to the user or society. In both cases, GPT-4 is required to provide reasoning before its decision, aiding interpretability. We sample responses with temperature 0. Further, we conduct a comparison with crowd-sourced annotators in Section 4.5, finding that in the cases of disagreements between GPT-4 and humans, errors in the human judgment, rather than GPT-4's analysis, cause the disagreement nearly 80% of the time. Complete prompts for GPT-4 evaluations can be found in Appendix A.1.

4.1 What Capabilities Arise from Scaling Pre-training vs Fine-tuning?

Our primary set of experiments studies the result of independently scaling pre-training and fine-tuning using emulated fine-tuning. For each dataset and model family, we generate responses to all 256 evaluation prompts using four models: a) the small fine-tuned model alone; b) the large fine-tuned model alone; c) the EFT *up-*

[3]Due to GPU memory constraints, we use Falcon-180B in 8bit inference mode when computing large-scale rewards for the Falcon down-scaling experiments, as both the 180B chat and base models cannot fit on 8 A100s in float16; quantization is likely to have some effect on generation quality. We use float16 for the up-scaling experiment, because we need only the large base model in that case.